



controls (one implementation each;
not compliance- or cost-matched)

	cells $R > 1$ Eq.2	worst margin	worst R
XPBD (tested)	8/24 9/24	+4e7 J	1e5
AVBD	2/24 3/24	+6.74 J	1.70
implicit	0/24 0/24	≈ 0	0.53

Scale gap, not a ranking: the tested XPBD host overdraws by up to 4.4×10^7 J, AVBD by at most 6.7 J (3/24 cells), the implicit realization never. E1: the AVBD/implicit accounting floor is $\leq 10^{-3}$ J, so 6.7 J is real injection, not noise.